

Introduction to Electrodynamics – David J. Griffiths

Notes by Gene

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1 Mathematical Review

1.1 Vector Analysis

A review of helpful vector stuff.

Scalar triple product: $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$.

- $\mathbf{B} \times \mathbf{C}$ is just the area spanned by $\mathbf{B}$ and $\mathbf{C}$, while $A \cos \theta$ is the altitude of $\mathbf{A}$ w.r.t. plane of $\mathbf{B}$ and $\mathbf{C}$
- Thus the scalar triple product is just the volume of the parallelepiped spanned by $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$.
- Consequently (order matters),

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

- You can swap dot and cross.

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$$

Vector triple product: Simplify using “BAC-CAB” rule.

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

With $\cdot$ and $\times$, there are 2 ways to construct a scalar (fg or $\mathbf{A} \cdot \mathbf{B}$) and 2 ways to construct a vector ($f\mathbf{A}$ or $\mathbf{A} \times \mathbf{B}$). This leads to 6 product rules.

1. $\nabla(fg) = f\nabla g + (\nabla f)g$
2. $\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
3. $\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
4. $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
5. $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
6. $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$

Quotient rules follow immediately from product rules.

Repeated application of the gradient, divergence, and curl result in further identities.

1. Divergence of gradient: the **Laplacian**

$$\nabla \cdot (\nabla T) \equiv \nabla^2 T = \frac{\partial^2}{\partial x^2} T_x + \frac{\partial^2}{\partial y^2} T_y + \frac{\partial^2}{\partial z^2} T_z$$

2. Curl of gradient: always 0.

$$\nabla \times (\nabla T) = 0$$

3. Gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$, not the Laplacian.

4. Divergence of curl: always 0.

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0$$

5. Curl of curl: use the following identity.

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}$$

1.2 Integral Calculus

Three important theorems:

1. Fundamental theorem of calculus for gradients:

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

2. The divergence theorem: Essentially “total flux $\rightarrow$ surface integral”

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}$$

3. Stokes’ theorem: Makes sense if you think “little curls within surface $\rightarrow$ integral over boundary”

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\partial S} \mathbf{v} \cdot d\mathbf{l}$$

1.3 Curvilinear Coordinates

For spherical coordinates, use θ as the polar angle (from z) and ϕ as the azimuthal angle (x - y plane).

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

- Infinitesimal line element: $d\mathbf{l} = (dr)\hat{r} + (r d\theta)\hat{\theta} + (r \sin \theta d\phi)\hat{\phi}$
- Infinitesimal volume element (Jacobian, anybody?): $dV = r^2 \sin \theta dr d\theta d\phi$

Gradient, divergence, and curl (which is a pain) are different for spherical coordinates. Consult Griffiths’ book (p. 42 or Appendix A) because it’s 4:30 am and I’m tired.

Cylindrical coordinates are easier. ϕ is the same as the azimuthal angle. z is the “height.”

$$x = s \cos \phi, \quad y = s \sin \phi, \quad z = z$$

- Infinitesimal line element: $d\mathbf{l} = (ds)\hat{s} + (s d\phi)\hat{\phi} + (dz)\hat{z}$
- Infinitesimal volume element: $dV = s ds d\phi dz$

1.4 The Dirac Delta

If we take the divergence of $\mathbf{v} = \hat{r}/r^2$, it’s 0. If we apply the divergence theorem,

$$\begin{aligned} \int_V (\nabla \cdot \mathbf{v}) d\tau &= \oint_S \mathbf{v} \cdot d\mathbf{a} \\ 0 &\stackrel{?}{=} \int \frac{1}{R^2} \hat{r} \cdot R^2 \sin \theta d\theta d\phi \hat{r} \\ 0 &\neq 4\pi \end{aligned}$$

The problem is that $\mathbf{v}$ does not have continuous partials. It blows up at $\mathbf{r} = 0$. To keep the divergence theorem consistent, we’ll later see:

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(\mathbf{r}) \longrightarrow \int (\nabla \cdot \mathbf{v}) d\tau = 4\pi$$

Dirac delta function in 1D: Technically a “distribution,” not a function. Defined as

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$
$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$$

The most important property, used over and over again, will be:

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a)$$

Holy measure theory, batman.

- We can also define a **step function** $\theta(x)$ such that

$$\delta(x) = \frac{d\theta}{dx}, \quad \theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

- The Dirac delta in 3D is just a repeated application of the Dirac delta in 1D. But there’s the interesting result of

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi\delta^3(\mathbf{r})$$

$$\text{Alternatively, } \nabla \cdot \left(\nabla \left(\frac{1}{r} \right) \right) = -4\pi\delta^3(\mathbf{r}) \longrightarrow \nabla^2 \frac{1}{r} = -4\pi\delta^3(\mathbf{r})$$

1.5 The Helmholtz Theorem

Maxwell’s equations treat $\mathbf{E}$ and $\mathbf{B}$ in terms of their divergence and curl.

Loosely, the **Helmholtz theorem** says that a “smooth” vector field with boundary conditions (e.g. decaying quickly as $r \rightarrow \infty$) can be uniquely defined in terms of the divergence-free component and a curl-free component.

For example, since the divergence of a curl is 0 and the curl of a gradient is 0, we can always reduce a field $\mathbf{F}$ to

$$\mathbf{F} = -\nabla\phi + \nabla \times \mathbf{A}$$

You may recognize these as the scalar and vector potentials.

2 Electric Fields

2.1 Key Equations

Coulomb’s law (for source charge q and test charge Q):

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}$$

$\epsilon_0 \approx 8.85 \times 10^{-12} C^2/(N \cdot m^2)$ is the permittivity of free space.

For multiple charges, we could simply say $\mathbf{F} = Q\mathbf{E}$ where the **electric field** $\mathbf{E}$ is defined by the charges

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{\mathbf{r}_i^2} \hat{\mathbf{r}}_i$$

Or for a continuous charge distribution (line, surface, volume, etc.)

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq_i}{\mathbf{r}_i^2} \hat{\mathbf{r}}_i$$

Just make sure to get the infinitesimal volume element, surface element, etc. right.

2.2 Gauss's Law

The flux of $\mathbf{E}$ through a surface S is a measure of the electric **field lines** passing through S .

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{a}$$

Since field lines emanating from, e.g. a positive charge within S must either intersect S or terminate in some negative charge within S , the flux through S is a measure of the charge within. Hence **Gauss's law**:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \equiv \Phi_E$$

- Requires a closed surface
- Heavily simplifies $\mathbf{E}$ calculations. Common Gaussian surfaces to use are the sphere, cylinder, and pillbox.

The differential form of Gauss's law follows from the divergence theorem:

$$\begin{aligned} \oint \mathbf{E} \cdot d\mathbf{a} &= \int_V (\nabla \cdot \mathbf{E}) dV \\ Q_{\text{enc}} &= \int_V \rho dV \\ \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \end{aligned}$$

There's also a neat result for the curl of $\mathbf{E}$. E.g. for a point source,

$$\begin{aligned} \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{\mathbf{r}^2} \hat{\mathbf{r}}, \quad d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}} \\ \oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} &= \int_S (\nabla \times \mathbf{E}) dS \\ \oint_{\partial S} \frac{1}{4\pi\epsilon_0} \frac{q}{\mathbf{r}^2} dr &= 0 = \int_S (\nabla \times \mathbf{E}) dS \\ \nabla \times \mathbf{E} &= 0 \end{aligned}$$

(For those of you who know Maxwell's equations – no, there's no conflict. This is the case of an electric field from static charges, not an electric field induced by a changing magnetic field.)

2.3 Electric Potential

Since $\nabla \times \mathbf{E} = 0$, by Stokes' we have $\oint \mathbf{E} \cdot d\mathbf{l} = 0$. We can see that the integral is independent of path. Thus we can define a “potential”:

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}$$

where $\mathcal{O}$ is the reference point. Potentials are most useful in terms of their difference:

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}$$

From the fundamental theorem of calculus for gradients,

$$\begin{aligned} - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} &= V(\mathbf{b}) - V(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\mathbf{l} \\ \mathbf{E} &= -\nabla V \end{aligned}$$

Example: The electric potential inside & outside a shell of charge with radius R . By Gauss's law,

$$\text{Inside: } \mathbf{E} = 0$$

$$\text{Outside: } \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$$

Now we can start to find the potentials. Outside, we have

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} = - \frac{1}{4\pi\epsilon_0} \int_{\infty}^{\mathbf{r}} \frac{Q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

Inside the shell, meanwhile,

$$\begin{aligned} V(\mathbf{r}) &= - \int_{\infty}^R \mathbf{E}_{\text{out}} \cdot d\mathbf{l} - \int_R^{\mathbf{r}} \mathbf{E}_{\text{in}} \cdot d\mathbf{l} \\ &= \frac{1}{4\pi\epsilon_0} \frac{Q}{R} - \int_R^{\mathbf{r}} 0 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \end{aligned}$$

Note that the potential is constant but not 0. Also note that the potential inside depends on the charges outside.

So far we have $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ and $\nabla \times \mathbf{E} = 0$. Since $\nabla \cdot \mathbf{E} = \nabla \cdot (-\nabla V) = -\nabla^2 V$, we recover **Poisson's equation**:

$$\nabla^2 V = - \frac{\rho}{\epsilon_0}$$

In the special case that there's no charge ($\rho = 0$), we have **Laplace's equation**:

$$\nabla^2 V = 0$$

2.4 Work and Energy in Electrostatics

Work in electrostatics:

$$W = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} = -Q \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = Q [V(\mathbf{b}) - V(\mathbf{a})]$$

$$W = Q\Delta V$$

The sign depends on whether we're acting along or against the direction of $\mathbf{E}$. Based on W , we can also see that **electric potential energy** is the minimum energy required to assemble a certain configuration.

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j>i}^N \frac{q_i q_j}{r_{ij}}$$

To get a formula for continuous charge distributions, note that

$$W = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j \neq i}^N \frac{q_i q_j}{r_{ij}} = \frac{1}{2} \sum_{i=1}^N q_i V(\mathbf{r})$$

The continuous-distribution case is simple enough, using $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$.

$$W = \frac{1}{2} \int V dq = \frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau$$

If we integrate by parts (go through details on this...) we get

$$W = \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla V) d\tau + \oint V \mathbf{E} \cdot d\mathbf{a} \right]$$

$$= \frac{\epsilon_0}{2} \left(\int_V E^2 d\tau + \oint_S V \mathbf{E} \cdot d\mathbf{a} \right)$$

At large distances, $E \propto 1/r^2$ and $V \propto 1/r$ while the surface grows as r^2 . By sending the whole volume integral to ∞ , we enclose all relevant charges while sending the surface integral to 0. Thus

$$W = \frac{\epsilon_0}{2} \int E^2 dV$$

The energy is stored in the $\mathbf{E}$ field with density $\frac{\epsilon_0}{2} E^2$ energy per unit volume.

2.5 Capacitors

In **insulators**, electrons can't wander too far from the atom. In **conductors**, electrons are free to move from atom to atom.

1. $\mathbf{E} = 0$ inside conductor: charges rearrange in least-energy configuration.
2. $\rho = 0$ inside conductor.
3. Net charges reside on surface, which is equipotential.

4. $\mathbf{E}$ just outside is perpendicular to surface.

We define the capacitance as $C = Q/V$, with V being the potential between two conductors.

1. E.g. for two parallel plates of area A and charges $-Q, +Q$, we have

$$\sigma = \frac{Q}{A}, \quad \mathbf{E} = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}, \quad V = \frac{Q}{A\epsilon_0}d$$

$$C = \frac{\epsilon_0 A}{d}$$

2. E.g. for two concentric spherical shells with radii a, b ; charge $+Q$ on inner and $-Q$ on the outer, we have

$$\text{Electric field between shells: } \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$V = - \int_a^b \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} \cdot d\mathbf{l} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$C = 4\pi\epsilon_0 \frac{ab}{(b-a)}$$

“Charging up” a capacitor means moving electrons from the positive plate to the negative plate. The total work necessary (and thus maximum potential of capacitor) is

$$W = \frac{Q^2}{2C} = \frac{1}{2}CV^2$$

This is derived from $dW = dqV$, so $\frac{dW}{dq} = q/C$ and integrate.

3 Electric Potentials

3.1 The Uniqueness Theorems (Laplace’s Equation)

Laplace’s equation $\nabla^2 V = 0$ is a special case of Poisson’s $\nabla^2 V = -\rho/\epsilon_0$ (make sure you can derive this from Maxwell’s equations).

1. Admits solutions called **harmonic functions**. Like “rubber sheets” that fit boundary as tightly as possible.
2. Value of B at (x, y, z) is average of values around (x, y, z) .

$$V(x, y, z) = \frac{1}{4\pi R^2} \oint_{\text{sphere}} V da$$

3. Extrema occur at boundaries (consequence of above)

3.1.1 1st Uniqueness Theorem

Uniqueness of potential w.r.t. **specified boundary or charge density + value of V on boundary.**

Statement: The solution to Laplace's equation in volume V is uniquely determined if V is specified on a boundary surface S .

Corollary: The potential in a volume V is uniquely determined if 1) charge density ρ throughout region, 2) value of V on boundaries, is given.

Proof: For (1), assume proof by contradiction with $\nabla^2 V_1 = 0$, $\nabla^2 V_2 = 0$, and $V_3 = V_1 - V_2$ that satisfies $\nabla^2 V_3 = 0$. Show that $V_3 = 0$. For (2), use same procedure with Poisson's:

$$\nabla^2 V_1 = -\frac{\rho}{\epsilon_0}, \quad \nabla^2 V_2 = -\frac{\rho}{\epsilon_0}$$

With this, we can prove a previous result.

Statement: The potential inside a conductor is constant, assuming there's no charge inside.

Proof: No charge, so $\rho = 0$ and $\nabla^2 V = 0$ inside the conductor. All charges are at the surface and distributed equipotentially, thus constant $V = V_0$ at surface. Extrema only at boundaries, thus $V = V_0$ everywhere.

3.1.2 2nd Uniqueness Theorem

Uniqueness of potential w.r.t. **total charge on each conductor.**

Statement: In a volume V surrounded by conductors and containing a charge density ρ , the electric field is uniquely determined if the **total charge** on each conductor is given.

Proof: Left as an exercise :d

3.2 The Method of Images

Useful to simplify problems involving charges and conducting planes (grounded).

Consider a plane in the xy -plane with a charge placed at $(0, 0, d)$. Essentially, we solve Poisson's equation under boundary conditions:

1. $V = 0$ when $z = 0$ (grounded conductor)
2. $V \rightarrow 0$ when $x^2 + y^2 + z^2 \gg d^2$

We consider the separate case where the charge is mirrored by the conductor. Then

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right] \quad (1)$$

Which solves the boundary conditions. By the uniqueness theorem, this solves the problem for $z \geq 0$.

- Note: for conducting plates, $E = \sigma/\epsilon_0$ (be able to explain how this differs with $E = \sigma/2\epsilon_0$). By comparison with (1), we see that the total induced charge $\int \sigma da = -q$, as expected.

- The charge $+q$ is attracted to the conducting plate with force

$$F = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z}$$

- The actual energy is half of what we get with images, though, since only the $z \geq 0$ region contributes. Recall that $W = \frac{\epsilon_0}{2} \int E^2 dV$.

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}$$

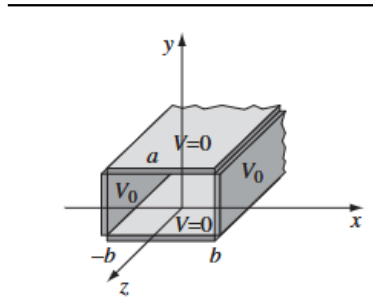
- For an example with spherical conductors, see p. 128.

3.3 Boundary Value Problems

3.3.1 Cartesian

E.g. consider two infinitely-long grounded plates. The boundary-value conditions are

- $V = 0$ at $y = 0, y = a$
- $V = V_0$ at $x = -b, x = b$



Problem setup (Griffiths 136)

The problem has z -independence. Assume a **separable solution** $V = X(x)Y(y)$ to Laplace's equation $\nabla^2 V = 0$.

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \quad \longrightarrow \quad \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0$$

By varying either x, y while holding the other constant, we see that both terms must be constant.

$$\begin{aligned} \frac{1}{X} \frac{d^2 X}{dx^2} &= k^2 \longrightarrow X(x) = Ae^{kx} + Be^{-kx} \\ \frac{1}{Y} \frac{d^2 Y}{dy^2} &= -k^2 \longrightarrow Y(y) = C \sin(ky) + D \cos(ky) \end{aligned}$$

- We use k^2 for ease later.
- Generally, if the dimension is bounded, use negative constant. If the dimension isn't bounded, use positive constant. It's a difference of exponential vs. sinusoid at end.

Symmetry in x dictates that $A = B$; absorb constant into C & D .

$$V(x, y) = (e^{kx} + e^{-kx}) (C \sin(ky) + D \cos(ky)) = \cosh(kx) (C \sin(ky) + D \cos(ky))$$

Now we know that $D = 0$ and $k = n\pi/a$ from remaining conditions. We now construct the general solution with boundary value conditions:

$$V(x, y) = \sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n\pi y}{a}\right)$$

Condition: $V(b, y) = \sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n\pi y}{a}\right) = V_0$

Fourier's trick: multiply by $\sin$ and integrate to exploit orthogonality of sines.

$$\sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi b}{a}\right) \int_0^a \sinh\left(\frac{n\pi y}{a}\right) \sinh\left(\frac{n'\pi y}{a}\right) dy = \int_0^a V_0 \sinh\left(\frac{n'\pi y}{a}\right) dy$$

$$\int_0^a \sinh\left(\frac{n\pi y}{a}\right) \sinh\left(\frac{n'\pi y}{a}\right) dy = \begin{cases} 0 & n' \neq n \\ \frac{a}{2} & n' = n \end{cases}$$

$$C_n \cosh\left(\frac{n\pi b}{a}\right) = \frac{2}{a} \int_0^a V_0 \sinh\left(\frac{n\pi y}{a}\right) dy = \frac{2V_0}{n\pi} (1 - \cos(n\pi)) = \begin{cases} 0 & n \text{ even} \\ \frac{4V_0}{n\pi} & n \text{ odd} \end{cases}$$

Finally, we plug into $V(x, y)$ to get:

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \frac{\cosh(n\pi x/a)}{\cosh(n\pi b/a)} \sin\left(\frac{n\pi y}{a}\right)$$

Though the formula is by no means general, the procedure is important to know.

3.3.2 Spherical (Azimuthal Symmetry)

Very similar to previous situation, but Laplace's equation is different.

$$\nabla^2 V = 0$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

Assume azimuthal symmetry and work with the separable solution $V = R(r)\Theta(\theta)$.

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0$$

Assume constants: $\begin{cases} \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = l(l+1) & \longrightarrow R(r) = Ar^l + \frac{B}{r^{l+1}} \\ \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -l(l+1) & \longrightarrow \Theta(\theta) = P_l(\cos \theta) \end{cases}$

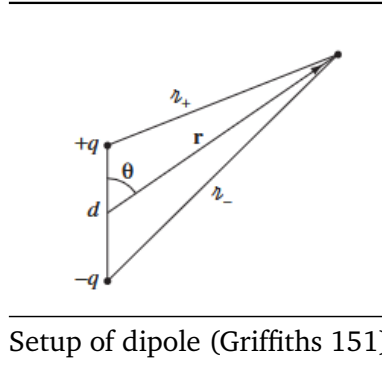
P_l are the **Legendre polynomials**. The overall solution is of the form

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(Ar^l + \frac{B}{r^{l+1}} \right) P_l(\cos \theta)$$

3.4 Multipole Expansion

3.4.1 Electric Dipoles

Far from a charge configuration, $V(r) \sim Q_{\text{tot}}/r$. For dipoles, $Q_{\text{tot}} = 0$, but we seek a more exact form of $V(r)$.



By the law of cosines,

$$r_{\pm}^2 = r^2 + (d/2)^2 \mp rd \cos \theta = r^2 \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right)$$

At $r \gg d$, the binomial expansion yields

$$\frac{1}{r_{\pm}} \approx \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right) \approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

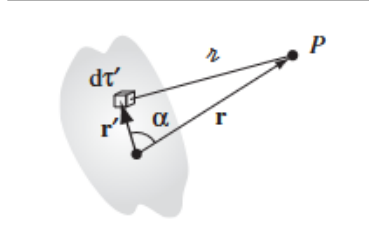
Thus $1/r_+ - 1/r_- \approx d \cos \theta / r^2$. The overall potential of the dipole at $r \gg d$ is approximately

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right) \approx \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

- Note that the potential goes like $\sim 1/r^2$, not $\sim 1/r$.
- This is a general feature. Quadrupoles go like $\sim 1/r^3$, octopoles go like $\sim 1/r^4$.

3.4.2 Multipole Expansion

Now we seek to find an exact form of $V(r)$ for an arbitrary charge distribution.



Problem setup (Griffiths 152)

The potential + law of cosines:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{z} \rho(\mathbf{r}') d\tau'$$

$$z^2 = r^2 + (r')^2 - 2rr' \cos \alpha = r^2 \left[1 + \left(\frac{r'}{r} \right)^2 - 2 \left(\frac{r'}{r} \right) \cos \alpha \right]$$

$$z = r\sqrt{1 + \epsilon}, \quad \epsilon \equiv \left(\frac{r'}{r} \right) \left(\frac{r'}{r} - 2 \cos \alpha \right)$$

As before, we binomial expand $1/z$ for small ϵ ($r \gg r'$). After plugging in ϵ and reorganizing, we find that

$$\frac{1}{z} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \alpha)$$

Thus

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \int (r')^n P_n(\cos \alpha) \rho(\mathbf{r}') d\tau'$$

- This is called “multipole expansion.” Note that the first term gives the monopole contribution, the second gives the dipole contribution, etc.
- Though this is exact, it’s best used in approximations.

3.4.3 Dipole Moment

For dipoles, the dominant term in the potential is

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int r' \cos \alpha \rho(\mathbf{r}') d\tau'$$

$$= \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}} \cdot \int \mathbf{r}' \rho(\mathbf{r}') d\tau'$$

We call the integral the **dipole moment**.

$$\mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') d\tau'$$

Note that for point charges, $\mathbf{p} = \sum q_i \mathbf{r}'_i$. Hence, similar to “center of mass” but more like “center of charge.” The dipole potential in terms of this is:

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$$

We can also find the electric field of a dipole. Set the center of the dipole at the origin with $\mathbf{p}$ in the $\hat{z}$ -direction.

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\hat{r} \cdot \mathbf{p}}{r^2} = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$$

Take gradients:

$$\begin{aligned} E_r &= -\frac{\partial V}{\partial r} = \frac{2p \cos \theta}{4\pi\epsilon_0 r^3} \\ E_\theta &= -\frac{1}{r} \frac{\partial V}{\partial \theta} = \frac{p \sin \theta}{4\pi\epsilon_0 r^3} \\ E_\phi &= -\frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} = 0 \end{aligned}$$

Thus the electric field of a dipole is

$$E_{\text{dip}}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

4 Electric Fields in Matter

4.1 Polarization

- **Conductors:** electrons free to move
- **Insulators/dielectrics:** electrons are “leashed” to atom

A neutral atom in E field can become **polarized**. E pulls nucleus & electron field apart, creating a dipole.

$$\mathbf{p} = \alpha \mathbf{E}, \quad \alpha = \text{atomic polarizability}$$

For molecules, the situation is more complicated. We have to use a **polarizability tensor**:

$$\begin{aligned} p_x &= \alpha_{xx} E_x + \alpha_{xy} E_y + \alpha_{xz} E_z \\ p_y &= \alpha_{yx} E_x + \alpha_{yy} E_y + \alpha_{yz} E_z \\ p_z &= \alpha_{zx} E_x + \alpha_{zy} E_y + \alpha_{zz} E_z \end{aligned}$$